

Safety Risk Assessment Based Minimum Separation Boundary for UAM Operations

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Abstract— Urban Air Mobility (UAM) is a promising next-generation mobility solution, leveraging airspace and state-of-the-art technologies for efficient and prompt transportation in urban areas. In UAM operations, constrained urban airspace and a high safety requirement level have presented new challenges in determining the separation boundary between aircraft; increasing airspace efficiency while maintaining safety is crucial. In general aviation and Unmanned Aerial Systems (UAS), separation boundaries are set to large values to readily meet a target level of safety (TLS). These large boundaries, however, can severely lower the traffic capacity of UAM airspace if applied directly. To address this issue, in this paper we propose a methodology for determining the safety-assured minimum separation boundary for UAM operations. Our methodology identifies system safety requirements, conducts safety risk assessments (SRA), and analyzes system integrity for a comprehensive systemic approach, which reduces the reliance on extensive flight data without compromising conservatism. We applied the proposed methodology to a Global Navigation Satellite System (GNSS)/Inertial Navigation System (INS) integrated navigation system as a case study for analysis. The results indicate that a significant reduction in separation boundaries is possible while still satisfying the safety requirements compared to existing separation boundaries for general aviation and UAS. Furthermore, this paper applies the proposed method to various navigation technologies such as multi-GNSS constellation, GNSS augmentation systems, and INS sensor with different grades with the goal of further reducing the separation boundary to satisfy the stringent vertical requirements of future UAM airspace.

Keywords— urban air mobility, separation boundary, target level of safety, safety risk assessment, integrity risk analysis

I. INTRODUCTION

Growing populations and restricted infrastructures in cities lead to transportation challenges in urban and suburban areas. As a promising next-generation mobility solution, Urban Air Mobility (UAM) leverages airspace and state-of-the-art technologies for efficient and prompt transportation in urban areas [1]. According to FAA's UAM ConOps, the UAM system coexists with National Air Space (NAS) and aims to operate on designated routes. As UAM systems reach maturity, conventional Air Traffic Control (ATC) will confront difficulties in managing increased traffic densities, necessitating an autonomous safety monitoring system. Therefore, autonomous Flight Path Management (FPM), ensuring vehicle separation and route adjustments, is essential for the safety of numerous UAM operations [2]. FPM conducts a deconfliction that predicts and resolves loss of separation with other traffic or static obstacles such as mountains and buildings. It establishes a separation boundary around the trajectory, verifying that no overlap occurs between the boundary and hazards.

This deconfliction effectively prevents most fatal accidents by setting separation boundaries in cross-track, along-track, and vertical directions. Due to system uncertainties, however, the separation boundary itself may fail to bound the aircraft's actual position, which is called a separation violation (SV) [3]. An SV can lead to unexpected mid-air collisions (MAC), resulting in serious casualties. Therefore, a separation boundary for which the probability of SV is lower than the risk requirement must be determined. In general aviation, large separation boundaries such as 5 nautical miles horizontally and 2,000 feet vertically [4] are implemented to readily meet the safety requirements related to the MAC rate. Such large boundaries are possible owing to the vastness of airspace. However, in UAM operations, these large separation boundaries could significantly lower airspace efficiency due to a markedly constrained airspace compared to general aviation [5]. Therefore, to enhance the traffic capacity of UAM, it is essential to determine a smaller separation boundary that can also satisfy the safety requirement.

Previous works [6 - 9] on determining the separation boundary for automated deconfliction [10] in general aviation did not derive the values based on the safety requirement. The separation boundary was instead derived from the trajectory prediction uncertainty by setting the bound on the maximum position error of commercial aircraft. The bound was determined from data collected by ground radar or estimated by a simulation. In other studies [11 - 13] for unmanned aircraft systems (UAS), the separation boundary, known as the well clear volume (WCV), is designed to reduce the rate of loss of separation while efficiently supporting the collision mitigation functions. Furthermore, those values are over a thousand feet, which is unsuitable for future UAM operations where the target value of vertical separation is approximately 250 ft [14]. Therefore, determining the minimum separation boundary could provide a practical starting point for UAM stakeholders to explore the effective use of airspace and establish an optimal separation boundary, particularly in the early phases of UAM implementation. The term "minimum" here implies the smallest allowable separation boundary. The "minimum" should not be confused with the "minima" found in *separation minima* [4], which means the separation boundary itself.

This paper proposes a new methodology for determining the safety-assured minimum separation boundary for UAM operations. We use a systemic approach to reduce reliance on extensive flight data. Our method involves identifying system safety requirements and allocating them throughout the UAM system, considering system uncertainties that can result in position errors and fatal accidents. The separation boundary protects the system against the position error due to system uncertainties. A risk assessment process is proposed

that bridges the safety requirement and system uncertainty. To practically implement the process without extensive data, we analyze the integrity risk of the system to conservatively evaluate the uncertainty in the position domain, accounting for the operational conditions of the system.

The structure of this paper is as follows: Section II outlines an overview of previous works on the separation boundary, Section III describes our methodology for determining the minimum separation boundary, Section IV presents a case study in which the proposed methodology is applied to a Global Navigation Satellite System (GNSS)/Inertial Navigation System (INS) integrated navigation system, and Section V concludes our paper with a brief summary of potential future works.

II. RELATED WORK

Previous works on the separation boundary [4, 15, 16] mainly focus on how to assess the safety of a given separation boundary. The Reduced Vertical Separation Minima (RVSM) [4], an airspace protocol that increases capacity by reducing vertical separation from 2000 ft to 1000 ft between flight levels FL290 and FL410, requires a safety assessment guided by ICAO to determine the risk of MAC. This assessment [15] entails collecting actual flight data with reduced separation and calculating the collision rate. For early UAM operation, however, RVSM is challenging to implement due to the lack of operational data, reference routes, and airspace designs.

Single European Sky Air Traffic Management (ATM) Research (SESAR) [16] presents a model with three barriers to mitigate the risk of MAC in UAS operations: strategic, tactical, and collision avoidance (CA) mitigations. Each of these barriers has an activation threshold (barrier) and evasive actions. The definition of the barriers is discussed in detail in [13]. Based on the SESAR barrier model, Building Basic Blocks for a U-Space Separation (BUBBLES) established a collision risk model which quantifies the probability of MAC by sequentially multiplying the failure probability of each barrier. The BUBBLES risk model provides a comprehensive method for analyzing collision risk based on the actions performed prior to a potential MAC. However, this method requires data from repeated simulations for a given separation boundary to calculate the failure probabilities [16]. In addition, the simulations, which assume only normal operational conditions, do not account for other operational conditions like sensor failure, which can substantially increase the risk of MAC.

The absence of flight data presents a new challenge for the early stages of UAM operations. Existing methods face limitations due to their reliance on operational data of a certain separation boundary and the assumption of normal operation. Consequently, data-driven assessment methods are unsuitable for deriving a minimum separation boundary satisfying the safety requirement. To address these limitations, this paper proposes a systemic approach with a high degree of conservatism to determine the safety-assured minimum separation boundary for UAM operations.

III. SAFETY ASSURED MINIMUM SEPARATION BOUNDARY

Our methodology involves three key elements: allocation of the target level of safety, safety risk assessment, and integrity risk analysis. The separation boundary has a

potential risk of SV which should be less than the SV risk requirement. The SV risk requirement is derived from the operation's target level of safety (TLS), expressed in units of fatal accidents per flight hour. As shown in Fig. 1, the TLS is allocated to subsystems whose uncertainties induce position errors, SV, and consecutively fatal accidents like a MAC. The subsystems could ensure the allocated TLS against their position error by establishing appropriate separation boundaries. Since not all SV leads to MAC, the fatality of the SV is evaluated to translate the allocated TLS to the SV risk requirement, denoted as $P_{SV,REQ}$. The safety risk assessment (SRA) in this paper calculates the fatality of SV under a UAM operation scenario and derives the SV risk requirement from the allocated TLS.

The minimum separation boundary is defined as the set of points at which the probability of a position error exceeding the separation boundary is equal to the SV risk requirement. While the position error has a zero-mean distribution under the nominal condition, a sensor failure results in a biased distribution. Here, the distribution can be estimated by utilizing the integrity risk analysis (IRA) method, which allows for a conservative estimation even in the absence of extensive flight data. IRA, a safety evaluation technique used in safety-critical systems, includes analyzing the risk of system failures and calculating the error distribution based on operational conditions such as sensor failures. This systemic approach to the separation boundary facilitates the calculation process by reducing the reliance on actual data while preserving conservatism.

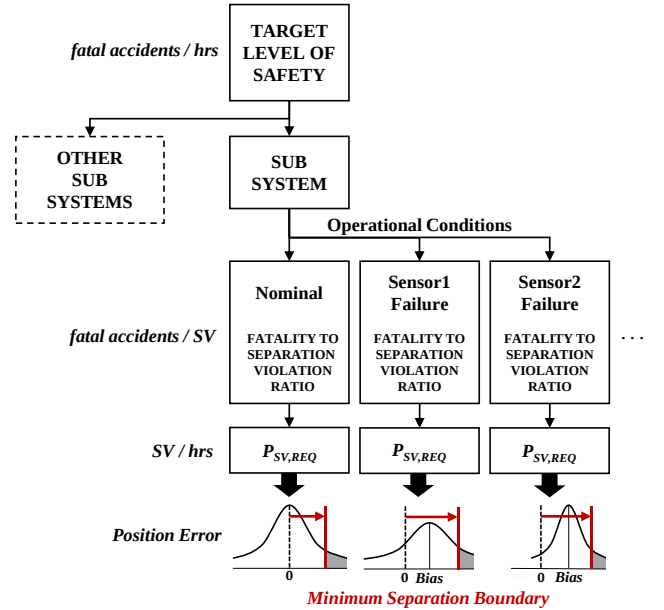


Fig. 1. The overview of the proposed methodology.

A. Target Level of Safety Allocation Scheme

TLS represents an acceptable operational risk in terms of fatal accidents per flight hour. After UAM systems including propulsion, navigation, and other subsystems are established, TLS is allocated to them. Each subsystem should be designed to meet its allocated TLS. The separation boundary is used as a design strategy for a specific subsystem to guarantee the allocated TLS by preventing fatal accidents caused by a position error known as the total system error (TSE). Because UAM is a safety-critical system, its TLS

may be as stringent as that of general aviation. To ensure a high degree of safety with the separation boundary, it is necessary to analyze the failure of subsystems with a high prior probability. In this paper, the navigation system is primarily analyzed because it is susceptible to sensor failures that may cause bias in the position error.

As described in Fig. 1, the TLS is allocated to each operational condition. Because the TSE distribution can vary with different operational conditions, fatality is assessed separately for each condition. The fatality-to-SV ratio translates the allocated TLS into the SV risk requirement. This ratio bridges the TLS and SV risk depending on the operation scenario. In this paper, we process the ratio as part of the SRA within the UAM operating scenario. The following subsection will discuss the SRA, a process that evaluates the potential risk of exceeding the separation boundary.

B. Safety Risk Assessment

We implement the SRA to assess the fatality of the separation boundary, which is used to convert the allocated TLS to the SV risk requirement. For a conservative SRA, our research established the fatality model quantifying the probability that SV results in a series of collisions and fatal accidents, represented as (1):

$$P(HH) = P(HH | CO)P(CO | SV)P(SV), \quad (1)$$

where $P(HH)$ is the probability of harm to humans (HH), $P(HH|CO)$ is the probability of HH given a collision (CO), $P(CO|SV)$ is the probability of CO during an SV, and $P(SV)$ is the probability of an SV. The allocated TLS is a requirement of $P(HH)$ and can be converted to the SV risk requirement using (1). $P(HH|CO)$ is assumed to be 1 when considering a high likelihood of human casualties resulting from a MAC. $P(CO|SV)$ represents the fatality-to-SV ratio after considering all procedures which will be performed to mitigate a potential MAC such as detect and avoid (DAA) systems [17, 18]. In this paper, however, $P(CO|SV)$ is calculated under the assumption that no mitigations are available. Then $P(CO|SV)$ becomes to depend only on traffic density, encounter geometry, and TSE distribution of engaged aircraft. This is because the development of collision mitigation for UAM operation is still ongoing, and even when completed, it requires a number of flight data to thoroughly compute their failure probability, as done for the BUBBLESE risk model [16].

CO can be defined as the overlap of two aircraft across all three axes [15]. Using this definition, (1) can be expressed as

$$P(CO | SV)P(SV) = P(CO, SV) = 2P_x(overlap, SV)P_y(overlap, SV)P_z(overlap, SV), \quad (2)$$

where the multiplier 2 is the number of aircraft involved in the collision, and $P(overlap, SV)$ is the probability of overlap between two aircraft with the subscript denoting the axis. To calculate (2) without an overlap probability derived from real aircraft of traffic data, we introduce the following worst-case scenario and virtual traffic. First, we assume that the collision can only be caused by the SV of the ownship ($P(CO|SV^C)=0$), implying that the virtual traffic is operating normally and no intrusion into the ownship's separation boundary occurs. Second, the ownship and traffic are in the

same corridor and comply with the same corridor requirements such as TLS and the separation boundary. Finally, the virtual traffic always exists on the same lateral and vertical plane of the ownship, and thus, the overlap probability in the along track direction, x , is always one ($P_x(overlap, SV)=1$). In detail, to assess the risk on lateral and vertical axes, we position the virtual traffic either right adjacent to the ownship ($P_x(overlap, SV)=1$) or directly below it ($P_y(overlap, SV)=1$), while maintaining separation from each other in the lateral or vertical direction. These probabilities are related to the traffic density and encounter geometry and could be replaced with values derived from future UAM traffic data and existing risk models [15, 19]. Fig. 2 illustrates the assumed scenario for vertical separation and can be easily extended to the case of lateral separation.

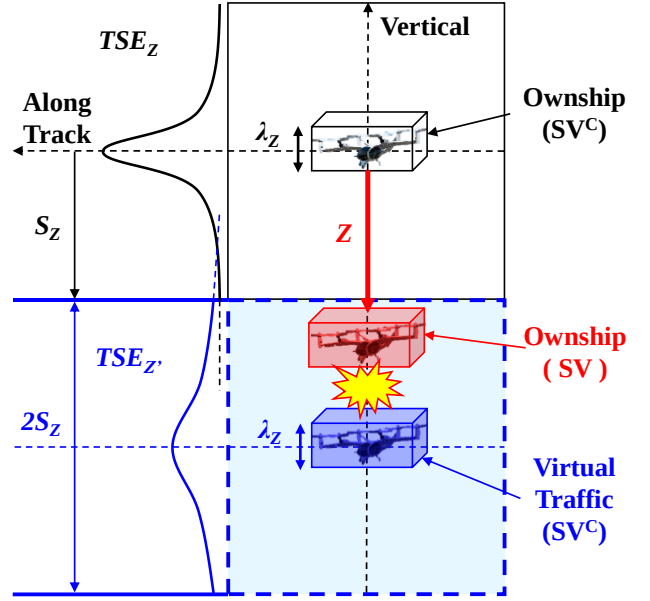


Fig. 2. Illustration of the vertical separation scenario. λ_z is the vertical dimensions of the aircraft bounding box, S_z is the vertical separation boundary, z is the position of the ownship and z' is the position of the traffic. TSE_z is the position error distribution of the ownship and $TSE_{z'}$ is the position error distribution of the traffic. When the ownship goes across its separation boundary ($S_z < z$) and overlaps with the normal traffic ($S_z < z' < 3S_z$), SV leads to an MAC.

For this scenario, the probability of collision can be expressed by (3) and (4) for lateral and vertical separations, respectively:

$$P(CO, SV_y) = 2P_y(overlap, SV_y) = 2 \int_{S_y}^{3S_y} \int_{y-2S_y-\lambda_y}^{y-2S_y+\lambda_y} g(y') f(y) dy' dy \quad (3)$$

$$P(CO, SV_z) = 2P_z(overlap, SV_z) = 2 \int_{S_z}^{3S_z} \int_{z-2S_z-\lambda_z}^{z-2S_z+\lambda_z} g(z') f(z) dz' dz. \quad (4)$$

In these equations, λ is the lateral or vertical length of the aircraft bounding box, S is the separation boundary with the subscript denoting the axis, $g(\cdot)$ is a probability density function of the traffic position error, and $f(\cdot)$ is the probability density function of the ownship position error. Each pdf of TSE distribution is assumed to be Gaussian. The outer integral interval is the range in which the traffic normally exists and the inner one is the overlap range. Equations (1) and (3, 4) compute the probability of fatal accidents which

must be lower than the allocated TLS. The proposed model is dependent on the separation boundary and the TSE of both the ownship and the traffic. If the TSEs are provided, the separation boundary corresponding to the allocated TLS can be derived from the model and the SV risk requirement is the probability of TSE exceeding the boundary.

In our SRA, the TSE stands as a key component. The TSE can be modeled as the zero-mean Gaussian with a variance of the root sum squared of the navigation system error (NSE) and flight technical error (FTE) as the path definition error (PDE) is considered negligible [20]. NSE is a position error caused by uncertainties in the navigation system, whereas FTE is a position error caused by uncertainties in the weather, pilotage, control system, and others. TSE distribution can be biased by abnormal NSE or FTE. NSE distribution of normal and failure conditions can be conservatively estimated through the IRA and is described in detail in the following subsection.

FTE in UAM operations is a relatively unexplored area, primarily due to the need for extensive flight data. Furthermore, detailed discussions regarding FTE are beyond the scope of this study. Instead, we utilize the concept of performance requirements to establish the maximum allowable value of FTE, as specified in performance-based navigation (PBN) [20]. In this regard, we assume that the navigation system is decoupled from other systems related to FTE, such as the weather, control and communication systems. Additionally, we consider the probability of simultaneous failure of these systems to be negligible when compared to the allocated TLS; thus, our analysis focuses on failures occurring in the navigation system. Therefore, FTE can be assumed to be a zero-mean Gaussian with the maximum allowable variance and combined into the resultant NSE to compute the TSE.

Unlike the ownship, quantifying the TSE of traffic can be challenging, especially when the state of the traffic is unknown. Similar to FTE, we derive the TSE of the virtual traffic from the corridor requirements. According to the UAM ConOps [1] and most UAM airspace concepts [5], only aircraft capable of maintaining a separation from other aircraft are authorized to operate in the corridor, which aligns with the concept of PBN, and is mandated as a corridor requirement. Therefore, the TSE of the traffic is derived from the corridor requirements, representing the maximum allowable value under the assumption that all traffic satisfies these requirements. This approach ensures a conservative estimate of collision risk. Following the definition of the accuracy requirement in PBN [20], which stipulates that the TSE of the aircraft must meet the accuracy requirement for 95% of its operation time, 2-sigma of the traffic's TSE is set to the given separation boundary.

We acknowledge that our model relies on several assumptions that may not align with all perspectives. However, these assumptions make the SRA more conservative favoring the aspects of safety. Although the approach may initially overestimate the risk, it remains a

viable and reliable strategy in the early stages of UAM, prioritizing safety above all else. As more traffic data on UAM operations becomes available, these assumptions can be progressively modified in a less conservative way.

C. Determining the Minimum Separation Boundary

The SRA discussed in the previous subsection is modeled to reduce the reliance on actual data. This subsection focuses on the conservative estimation of the ownship's TSE, a primary component in the SRA, through the IRA. Then, the subsection concludes with the determination of the minimum separation boundary. However, before performing the IRA, a baseline navigation system must be established, allowing IRA to quantify the TSE. In this study, the baseline navigation system is a GNSS/INS integrated navigation system, which serves as a basis for navigation systems with multiple sensors.

Fig. 3 illustrates how the proposed methodology is applied to the GNSS/INS integrated navigation system. Following the solid arrow, the TLS is directly allocated to the vertical positioning of the navigation system and each operational condition (i.e. Nominal, GNSS Failure, INS Failure). The allocated TLS and SRA are then used to determine the minimum separation boundary for each condition.

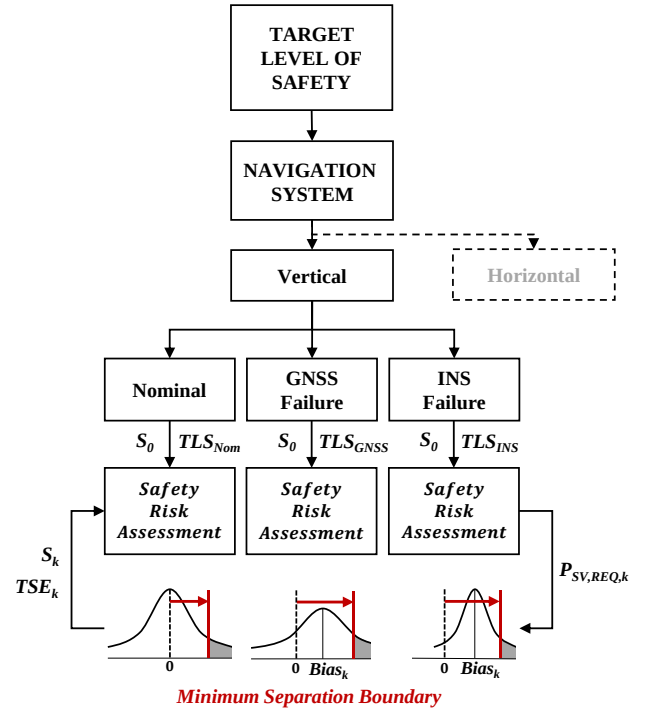


Fig. 3. Illustration of the proposed methodology applied to the GNSS/INS integrated navigation system. TLS_{Nom} , TLS_{GNSS} , and TLS_{INS} are the allocated TLS of each condition and S_0 is the initial separation boundary for the iteration. S_k is the separation boundary at the k -th epoch of the iteration, $P_{SV,REQ,k}$ is the SV risk requirement, $bias_k$ is the ownship's NSE bias and TSE_k is the ownship's TSE. The minimum separation boundary is the point where the separation boundary converges.

In general aviation, an IRA assesses the reliability of the navigation system for safety-critical operations such as precision approaches. The protection level (PL), which parallels the concept of the separation boundary, is computed using the IRA. This process ensures a high level of safety

against integrity risk (IR), which is the probability of a position error exceeding a predefined limit without alerting the user. The IRA uses integrity monitors to alert users of a deviation due to exceeding the limit of system uncertainties, particularly from sensor failures, and quantifies the undetected errors corresponding to the integrity risk requirement.

In this paper, we assume that UAM operation, which is safety-critical, employs integrity monitors. These monitors alert whenever a sensor failure is detected and subsequently abort operations. Therefore, IR is the SV risk in this system, implying that the integrity risk requirement is the SV risk requirement. The undetected fault, which is an NSE bias, is derived from the SV risk requirement using the IRA. More detailed information on this can be found in the referenced studies [21, 22]. As a result, the resulting separation boundary based on TSE obtained from the IRA is considered to be conservative.

As shown at the bottom of Fig.3, the numerical iteration is used to determine the minimum separation boundary. This is because there is no analytic solution for the separation boundary in (3, 4) and the SRA depends on the TSE which in turn depends on the outcome of the SRA: the SV risk requirement. Initially, a large separation boundary denoted as S_0 is assumed, which leads to a lower fatality-to-SV ratio and an increased SV risk requirement, subsequently resulting in a smaller separation boundary in the next iteration. This iterative process continues until it reaches convergence, progressively reducing the separation boundary. The point of convergence is the minimum separation boundary.

Because the distribution of TSE varies depending on the operational condition, the minimum separation boundary is calculated for each condition. Among the boundaries, the largest one is determined as the final minimum separation boundary for UAM operations. This minimum separation boundary can ensure safety across all conditions, and is derived from (5) as

$$S_{min} = \max(S_{Nom}, S_{GPSF}, S_{IMUF}), \quad (5)$$

where S_{min} is the final minimum separation boundary and S is the minimum separation boundary with the subscript denoting the operational condition.

While failure conditions can increase the separation boundary as a result of their bias, it is important to note that the low prior probability of such conditions adjusts the separation boundary by assigning a small weight to the fatality. Therefore, these separation boundaries become comparable to that of the normal condition. The relation can be expressed as follows:

$$\begin{aligned} TLS_{Nominal} &= P(HH | Nom) \cdot P(Nom) \\ TLS_{GPSF} &= P(HH | GPSF) \cdot P(GPSF) \\ TLS_{IMUF} &= P(HH | IMUF) \cdot P(IMUF), \end{aligned} \quad (6)$$

where $TLS_{Nominal}$, TLS_{GPSF} , and TLS_{IMUF} are the allocated TLS of each condition, $P(Nom)$, $P(GPSF)$, and $P(IMUF)$ are the prior probability of each condition, and $P(HH|Nom)$, $P(HH|GPSF)$, and $P(HH|IMUF)$ are the fatality of each condition.

IV. CASE STUDY ; GNSS/INS INTEGRATED NAVIGATION SYSTEM

A. Simulation Conditions

In this section, the proposed methodology is applied to a GNSS/INS integrated navigation system as a case study. The case study assumes that an ownship is equipped with an extended kalman filter (EKF) based GNSS/INS integrated navigation system. As in Fig. 3, the sensor fault condition of the GNSS/INS integrated navigation system can be classified as fault-free (nominal), GNSS failure (GPSF), or INS failure (IMUF) hypothesis. We disregard GNSS/INS concurrent failures as their possibility is negligible compared to TLS. For simplicity, the TLS of the navigation system in vertical direction is equally allocated to the three hypotheses.

NSE variance is the position error variance estimated by EKF. The NSE bias due to sensor failures is calculated through the IRA, as discussed in Section III.C. Particularly, two fault monitors are implemented: a residual-based receiver autonomous integrity monitoring (RAIM) [21] for GNSS failure and an innovation-based RAIM [22] for INS failure. The prior probability of GPSF or IMUF is 10^{-4} . For fault monitors, each threshold is defined based on an allocated continuity risk requirement [21]. Both biases are then determined by the minimum detectable error (MDE), which is a bias of NSE whose probability of being below the threshold (missed detection) matches the SV risk requirement. Finally, the TSE of the ownship is the root sum squared of the FTE requirement and the estimated NSE from the EKF and the IRA. For the EKF, a standard GPS (single constellation and single frequency) and consumer grade inertial measurement unit (IMU) are used. The detailed parameters are shown in [22, 23]. Table I summarizes the parameters utilized to estimate the TSE of the ownship, as well as the estimated variance of NSE and TSE.

TABLE I. SPECIFICATIONS OF THE OWNSHIP FOR THE CASE STUDY

Parameter	Value
Navigation System	GNSS/INS EKF
Fault Monitor	GNSS Fault - Residual Based RAIM
	INS Fault - Innovation Based RAIM
Fault Bias	Minimum Detectable Error
$TLS_{Nominal}$ TLS_{GPSF} TLS_{IMUF}	$TLS_{NAV,Vert} / 3$
σ_{NSE}	26.93 m
σ_{FTE}	15 m
σ_{TSE}	30.83 m
Size (Height)	6 m

TABLE II. CORRIDOR REQUIREMENTS FOR THE CASE STUDY

Requirements	Value
$TLS_{NAV,Vert}$	$10^{-6} \sim 10^{-10}$ fatal accidents/hrs
$P_{Continuity Risk}$	$2 \cdot 10^{-6}$ outages/hrs
95% Accuracy ($2\sigma_{TSE}$)	S_k
σ_{FTE}	15 m
Max. Size (Height)	6 m

In this study, we introduced the concept of corridor requirements, setting some parameters to their maximum allowable values. Table II outlines these corridor requirements, which are applied to both the ownship and traffic. The TLS of the navigation system in vertical direction is between 10^{-6} and 10^{-10} . The continuity risk requirement is $2 \cdot 10^{-6}$, which is allocated equally for GPSF and IMUF. The 95% accuracy requirement is the separation boundary at the k -th epoch. FTE is 15 m, in accordance with the FTE requirement of level flight below 1,500 m in PBN [20]. The maximum aircraft height is 6 m, referencing the height of the heavy-lifting helicopter, CH-47. The TSE of the traffic is the zero-mean Gaussian with variance equal to the accuracy requirement, and the height of the ownship and traffic is the size requirement. The following subsection covers the results of our case study.

B. Results from Case Study

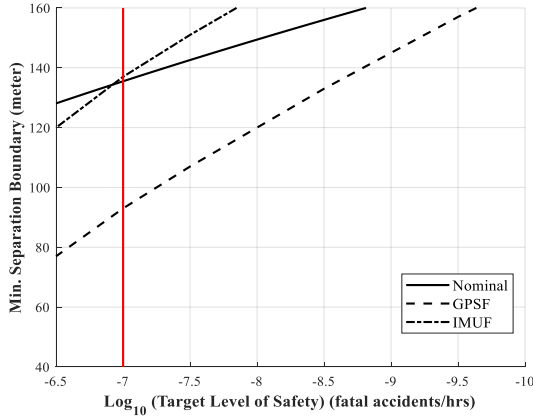


Fig. 4. Vertical separation boundary with allocated TLS for different operational conditions.

The results from the case study are shown in Fig. 4, which presents the minimum separation boundary in vertical direction for each sensor fault condition versus the TLS allocated to the hypothesis. Here, the x-axis is arranged in a reversed order for clarity and a smaller TLS implies a more

stringent safety requirement. The separation boundary increases as the allocated TLS decreases. For a given TLS of 10^{-7} , the IMUF curve has the largest separation boundary and the GPSF curve presents the smallest; however, this could vary depending on the prior probability for each hypothesis. For an allocated TLS of 10^{-7} , the separation boundaries of the nominal, GPSF, and IMUF are about 135 m, 93 m, and 137 m, respectively. In this case, the TSE variance for all conditions is 30.83 m which is the root sum squared of FTE (15 m) and NSE (26.93 m). The GPSF has a TSE bias of 36 m and a $P_{SV,REQ}$ of 0.0322, while the IMUF has a TSE bias of 95 m and a $P_{SV,REQ}$ of 0.0869. Consequently, the final vertical minimum separation boundary in this case study was set to 137m, which is determined by the IMUF.

Despite the significant reduction from previous standards of general aviation [4], a separation boundary of 137 m may be deemed insufficient for future UAM operations, given the vertical separation target value of 250 ft (76.2 m) [14]. This implies that the vertical separation boundary may be reduced even further. There are two approaches to address this issue within the scope of this paper: improving the navigation performance of the ownship or establishing stringent corridor requirements to reduce the traffic TSE. The following subsection will focus on the first approach and discuss the effect of the navigation performance of the ownship on the separation boundary.

C. Sensitivity to Navigation System Performance

Improving the navigation performance of the ownship provides significant benefits, such as reducing the ownship's TSE and enhancing robustness against sensor failure. These benefits, in turn, can contribute to reducing the minimum separation boundary. In this subsection, we conduct sensitivity studies by varying GNSS/INS system candidates, including multi-constellation (MC), dual-frequency (DF), tactical grade (TG) high-performance INS [23] and satellite-based augmentation system (SBAS) [24], to further understand their impact on the minimum separation boundary.

Multi-constellation can improve the accuracy of the position solution by increasing the number of range measurements and improving satellite geometry. Utilizing dual-frequency can improve the accuracy of the position solution because of an increased number of range measurements. High-performance INS, being more accurate, can enhance the position solution. Lastly, the SBAS can enhance the accuracy of the position solution based on correction information. In this case study, the benefit from GNSS monitoring capability of the SBAS is not considered. The EKF is applied to each candidate as before to estimate their TSE variance. Each candidate and its corresponding $\sigma_{TSE,Ownship}$ are detailed in Table III.

TABLE III. COMPARISON OF CANDIDATE NAVIGATION TECHNOLOGIES

Parameter	I(Ref.)	II	III	IV	V
Constellation	GPS	GPS+GAL	GPS	GPS	GPS
Frequency	L1	L1	L1+L2	L1	L1
IMU Grade	Consumer	Consumer	Consumer	Tactical	Consumer
SBAS	X	X	X	X	O
$\sigma_{TSE,Ownship}$ (m)	30.83	23.59	21.36	16.91	15.30

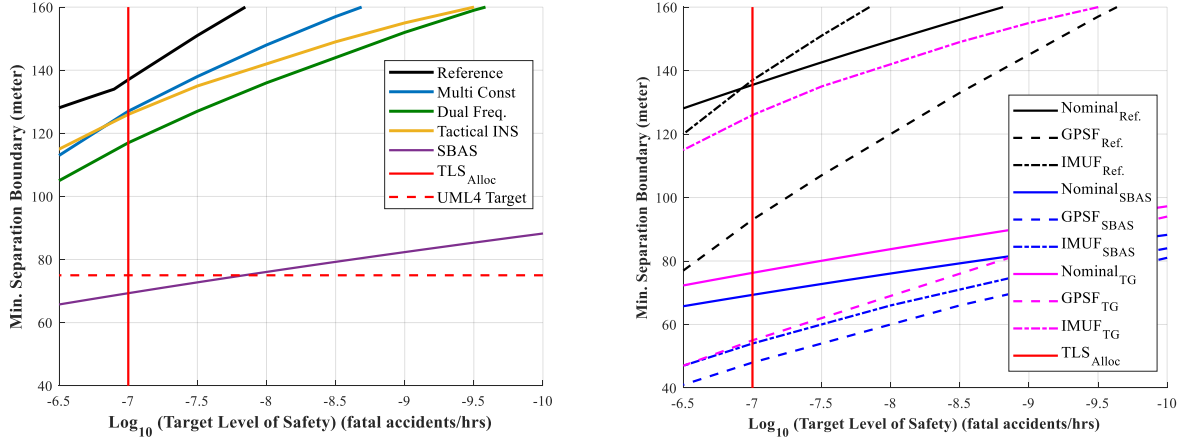


Fig. 5. Comparative analysis of minimum separation boundaries across GNSS/INS system candidates (left) and detailed view of the separation boundary for TG and SBAS (right).

The left side of Fig. 5 shows the final minimum separation boundaries of each candidate in comparison to the reference from the previous subsection (black line). Here, the x-axis is arranged in reversed order. All candidates show a reduction of the minimum separation boundary. In this case, the SBAS has a noticeable reduction, achieving a vertical minimum separation boundary of 69 m, exhibiting a reduction of about 50% compared to the reference value. This is due to its superior position accuracy and robustness against all sensor failures. However, high-performance INS, whose TSE is comparable to that of the SBAS, is less effective. As depicted on the right side of Fig. 5, this is because high-performance INS increases the navigation system's reliance on the INS making it vulnerable to IMU failures, which results in a larger separation boundary under IMUF conditions. Consequently, the GNSS/INS integrated system which utilizes SBAS corrections emerges as the most effective candidate for reducing separation boundaries. In addition, we recommend considering a combination of these technologies in the goal of further reducing the minimum separation boundary.

V. CONCLUSION AND FUTURE WORK

In UAM operations, constrained airspace and a high level of safety requirements have presented a new challenge in determining the size of the separation boundary. In this paper, a methodology was proposed for determining the safety-assured minimum separation boundary for UAM operations. Utilizing the TLS allocation scheme, safety risk assessment, and integrity risk analysis, the minimum separation boundary was derived in a conservative way, even in the absence of extensive data. The methodology was applied to a KF-based GNSS/INS integrated navigation system as a case study. The minimum separation boundary, which is significantly lower than conventional values while satisfying the safety requirements, was determined. As an extended case study, various GNSS/INS system candidates were analyzed to further reduce the vertical minimum separation boundary. Our results suggest that using SBAS corrections, showing up to a 50% improvement, is the most effective method, satisfying the stringent vertical requirements of future UAM airspace. Ultimately, the method developed in this study can provide a practical starting point for UAM stakeholders to explore the effective use of airspace and establish an optimal

separation boundary, particularly in the early phases of UAM implementation.

To further expand the scope of this research, future work can focus on addressing the conservative assumptions used in this research:

- Considering collision mitigation can reduce the minimum separation boundary by lowering the fatality of SV. It is noted that, once any collision mitigation with an activation threshold is considered, the definition of separation boundary is changed to the threshold.
- Considering the traffic scenario can reduce the minimum separation boundary by lowering the fatality of SV. To obtain the overlap probability through the data-driven approach, UAM traffic data should be carefully collected and classified with each flight mode of UAM operations such as level flight, climb/descend and encounter geometry.

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